

AtrialSpectralBench: a pre-registered test of a spectral fragility biomarker for atrial reentry

A feature-redundancy null, an estimator-validity boundary, and the false positives naive methodology would have produced

Mounish Mavuduru*

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Abstract

We ask whether a closed-form spectral graph biomarker predicts reentry inducibility in atrial tissue beyond established fibrosis and connectivity features. The biomarker is the **Spectral Fragility Index (SFI)**, the Fiedler-value sensitivity $\partial\lambda_2/\partial w_{ij} = (\varphi_i - \varphi_j)^2$ aggregated under a fibrosis-weighted conduction-uncoupling field. Pre-registered before any labels were seen, the primary endpoint was $\Delta\text{AUC} \geq 0.05$ with a DeLong paired $p < 0.05$ on shape-family-held-out folds. *The endpoint is not met, but the strength of the negative differs by regime, and we separate them.* On up to 100,000 synthetic excitable networks SFI adds no practically meaningful predictive value (subspace $\Delta\text{AUC} \approx +0.003$; $\sim 4,150$ cases needed at 80% power, resolvable only because N is huge, and so *clinically undetectable* there). On two real cohorts (182 patients: Roney $n = 100$, UW/Boyle $n = 82$) the combined gradient-boosting estimate is $+0.012$ with a bootstrap interval $[-0.017, +0.044]$ that excludes the threshold, so the real-anatomy result is a measured null rather than a shortfall of power. An earlier version of this work reported $+0.051$ here and called the real-cohort evidence inconclusive. That figure was an artefact of a defect in one cohort’s released substrate, which we found only afterwards; the audit that found it, and the three controls that failed to explain everything it left behind, are reported in full. Where the null is clean, the mechanism is feature redundancy: the exact per-region $\Delta\lambda_2$ SFI, which has no first-order approximation error, also adds nothing over the competitor set. A separate, label-free result explains why the *single-vector* SFI is also ill-conditioned on real tissue. It is valid only while $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2) = O(1)$, and real atria have $\rho \approx 2422$ ($\sim 10^3 \times$ past the empirical $\rho^* \approx 3$), so by Davis–Kahan the Fiedler vector is unstable. We do *not* claim ρ proves non-predictiveness; it bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a ranking feature. The spectral gap collapses as $N^{-1.05}$ on the real atrial surface (the 2-manifold Weyl exponent), so ρ grows lawfully with resolution. One component partially survives: the Fiedler-gradient localizer $|\nabla\varphi_2|$ is a weak but real rank localizer (origin in the top $\sim 18\%$) that beats graph-centrality baselines and transfers across three dynamical media, though it *coincides with*, and does not beat, substrate imaging, by a mechanism we make precise. Naive methodology (random CV, p -value thresholds, no effect-size gate) would have reported SFI as working; our pre-registered protocol rejects it. Our own convergence study surfaces a key limitation: the inducibility labeller over-calls at the operating mesh resolution and, on a completed six-tier study (1500–48000 nodes, $n = 24$), does not converge within the tested window. Our contribution is a carefully bounded negative result: the mechanism behind it, a label-free criterion for the estimator’s validity, a characterization of how it scales, and a demonstration of the false positives that standard methodology produces.

Keywords: algebraic connectivity; Fiedler value; spectral graph theory; atrial fibrillation; reentry; biomarker; perturbation theory; validity radius; pre-registration; negative result.

*Independent, unaffiliated in-silico computational-cardiology project. Correspondence and full source: see the project repository.

This is a 100% in-silico computational study. Ground-truth labels are electrophysiology-simulator inducibility verdicts, never clinical outcomes. All numbers trace to `results/*.json` and `notebooks/lab_notebook.md` in the project repository.

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1 Introduction

Algebraic connectivity λ_2 (Fiedler 1973 [1]) governs the stability of coupled excitable and oscillator networks (the master-stability-function result [3] and the synchronizability eigenratio λ_N/λ_2 [4]), and has been used as a real-time seizure detector [5]. The per-edge sensitivity

$$\frac{\partial \lambda_2}{\partial w_{ij}} = (\varphi_{2,i} - \varphi_{2,j})^2 \quad (1)$$

is a classical identity underlying Fiedler-vector edge-editing heuristics [2]. It is therefore natural to hypothesize that aggregating this sensitivity over a fibrosis-weighted perioperative uncoupling field yields a per-region biomarker of atrial-fibrillation (AF) reentry fragility. We test this hypothesis and *pre-register the null* (“SFI merely re-encodes fibrosis”) as an accepted, reportable outcome, so that a negative result is a success of the protocol rather than a file-drawer casualty.

The mathematics we use is classical; we claim no new theorem. Our contributions are (i) a leakage-controlled incremental test of SFI against real competitors and a spatial null on real anatomy; (ii) a validity criterion and a Weyl scaling law that explain the failure and predict it for the whole class of spectral fragility biomarkers; (iii) a cross-medium transfer analysis isolating the one component that generalizes; and (iv) a falsification-protocol demonstration showing that standard methodology would have manufactured exactly the false positive this protocol catches.

In brief, a closed-form spectral fragility index does *not* add practically meaningful reentry-inducibility prediction beyond existing features, either at synthetic scale or on 182 patients of real anatomy, and the mechanism is feature redundancy. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). What we report is not a discovery but a carefully drawn boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how readily naive methodology mistakes redundant, inconclusive signal for a working biomarker.

2 Theory: the Spectral Fragility Index

The core mathematics is classical numerical linear algebra; the contribution is its *use*. We state each result with its prior art (Section 2.5) and make the validity boundary explicit (Section 2.4).

2.1 The weighted conduction graph

Nodes are mesh vertices; edges are triangle edges; the edge weight $w_{ij} > 0$ is a fibre-anisotropic conduction conductance attenuated by local fibrosis. The combinatorial Laplacian is

$$L = D - W, \quad D = \text{diag} \left(\sum_j w_{ij} \right). \quad (2)$$

L is symmetric positive-semidefinite with a single zero eigenvalue (the constant vector) for a connected graph, with spectrum $0 = \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$ and eigenvectors $\varphi_1, \varphi_2, \dots$; λ_2 is the algebraic connectivity and φ_2 the Fiedler vector.

2.2 The closed-form fragility (first order)

L is a sum of rank-one edge terms, so

$$L = \sum_{(i,j) \in E} w_{ij} (e_i - e_j)(e_i - e_j)^\top \implies \frac{\partial L}{\partial w_{ij}} = (e_i - e_j)(e_i - e_j)^\top. \quad (3)$$

For a *simple* eigenvalue λ with unit eigenvector φ , first-order (Hellmann–Feynman) perturbation theory gives $\partial \lambda / \partial w_{ij} = \varphi^\top (\partial L / \partial w_{ij}) \varphi = (\varphi_i - \varphi_j)^2$, which for the Fiedler pair is Eq. (1): a non-negative per-edge *fragility*. Reducing a conductance w_{ij} by a small $\Delta w_{ij} \geq 0$ (uncoupling) lowers λ_2 by $\Delta w_{ij} (\varphi_{2,i} - \varphi_{2,j})^2$ to first order.

The identity is verified in seven independent ways rather than merely asserted. Computer algebra: sympy proves Eqs. (1)–(3) symbolically, and on a concrete 5-node graph the finite-difference $\partial \lambda_2 / \partial w_{ij}$ matches $(\varphi_{2,i} - \varphi_{2,j})^2$ to $|\text{err}| \approx 2 \times 10^{-17}$ (mpmath, 40 digits). Exactness: the identity holds *exactly* with rational weights (zero floating point); an interval-arithmetic enclosure certifies the true λ_2 to within $\sim 4 \times 10^{-40}$; and λ_2 / φ_2 agree across scipy, networkx, and a hand-rolled inverse iteration (spread 2.5×10^{-15}). The algebraic core is kernel-checked in the Coq proof assistant over the exact rationals, axiom-free (Appendix A). The analytic Hellmann–Feynman step (differentiability of a simple eigenvalue) is the cited classical result [2], not re-proved here.

2.3 Per-region aggregation: the SFI

Model diffuse perioperative stress as a distribution of small, non-negative, fibrosis-weighted fractional conductance reductions Δw (mean $\mathbb{E}[\Delta w_{ij}] = \text{frac}_{ij} w_{ij}$, calibrated to literature perioperative conduction-velocity slowing; $\Delta w \approx 0.36$, frozen in the pre-registration). The per-region Spectral Fragility Index is the expected λ_2 drop contributed by region R :

$$\text{SFI}(R) = \mathbb{E}[\Delta\lambda_2 \text{ from } R] \approx \underbrace{\sum_{(i,j) \in R} \mathbb{E}[\Delta w_{ij}] (\varphi_{2,i} - \varphi_{2,j})^2}_{\text{first order}} + \underbrace{\sum_{k \neq 2} \frac{(\varphi_k^\top \Delta L \varphi_2)^2}{\lambda_2 - \lambda_k}}_{\text{2nd-order resolvent}} + O(\|\Delta L\|^3). \quad (4)$$

The second-order term is the source of the validity radius below. Near eigenvalue degeneracy ($\lambda_2 \approx \lambda_3$) the single φ_2 is an arbitrary rotation in the low invariant subspace, so we use a basis-independent *subspace (projector) SFI*: with $P = \sum_c \varphi_c \varphi_c^\top$ over the clustered subspace, $s_{ij}^\top P s_{ij} = \sum_c (\varphi_{c,i} - \varphi_{c,j})^2$ ($s_{ij} = e_i - e_j$) is well-defined. For discrete edits (Maze lesion, atriotomy) the expansion is abandoned and λ_2 is recomputed *exactly* after the rank-one edit.

2.4 The validity radius (the central caveat)

The first-order form is trustworthy only while the perturbation is small *relative to the spectral gap* $\lambda_3 - \lambda_2$. Two different tools bound two different things (a common error is to use Weyl for both):

- Weyl’s inequality bounds the *eigenvalue* move: $|\Delta\lambda_2| \leq \|\Delta L\|$.
- Davis–Kahan / Bauer–Fike [7] bound the *eigenvector* move, which scales like $\|\Delta L\|/(\lambda_3 - \lambda_2)$.

This, not Weyl, governs when φ_2 (hence the single-vector SFI) degrades.

Define the *validity radius*

$$\rho := \frac{\|\Delta L\|}{\lambda_3 - \lambda_2}. \quad (5)$$

From second-order Rayleigh–Schrödinger theory the relative error of the first-order $\Delta\lambda_2$ prediction is $\Theta(\rho)$, and the Fiedler vector rotates by $\sin \angle(\varphi_2, \varphi'_2) \leq \sqrt{2} \rho$ (Davis–Kahan). Empirically the error is 1% at $\rho = 0.3$, crosses 10% at $\rho^* \approx 3$, and reaches 92% at $\rho = 28$, with Weyl holding throughout. On real excitable media the gap is tiny, so ρ is enormous: atrial-cohort median $\rho \approx 2422$, about $10^3 \times$ past ρ^* .

The criterion ρ proves that the *single-vector* first-order SFI is a numerically *invalid estimator* of $\Delta\lambda_2$ on real tissue, and that φ_2 is ill-conditioned (Davis–Kahan), a label-free statement about the estimator. It does *not* prove SFI is non-predictive: ρ bounds the accuracy of the $\Delta\lambda_2$ *magnitude*, not the classification content of a monotone ranking feature; indeed the *exact* $\Delta\lambda_2$ SFI (no ρ limitation) is *also* non-predictive (Section 4.1), so the predictive null is separately due to feature redundancy. Two caveats: (i) ρ ’s numerator carries the fibrosis-weighted Δw field, so only its denominator $\lambda_3 - \lambda_2$ is truly label-free; (ii) $\rho^* \approx 3$ is calibrated on a 16-node path graph with a problem-dependent constant, so “ $10^3 \times$ past ρ^* ” is order-of-magnitude, not exact.

2.5 Prior art (this is not new mathematics)

The identity (1) is Ghosh & Boyd (2006) [2]; algebraic connectivity is Fiedler (1973) [1]; λ_2 governing coupled excitable/oscillator stability is Pecora & Carroll (1998) [3] with the eigenratio of Barahona & Pecora (2002) [4]; Weyl’s law, Davis–Kahan, and second-order perturbation theory are classical (1970s and earlier). No result in Section 2 is a new theorem, and none is presented as one. The contribution is recognizing that ρ is the right *a priori* validity screen for a spectral fragility biomarker, measuring its violation on real tissue, and using it to *mechanistically explain* a negative result.

3 Methods

Full detail is in the pre-registration (`docs/PRE_REGISTRATION.md`) and theory note (`docs/SFI_THEORY.md`); the endpoints, cross-validation scheme, baseline set, and accepted null were frozen before any labels were inspected.

Substrate. Real left-atrial surfaces are loaded into weighted conduction graphs from two public cohorts: the Roney LA virtual cohort (Zenodo 5801337 [11]) and the UW/Boyle LGE-MRI cohort (Dryad 10.5061/dryad.kkwh70sg0, 82 distinct AF patients [12]). Each mesh carries universal atrial coordinates, an image-intensity-ratio (IIR) fibrosis proxy mapped to a fibrosis field by the Khurram healthy-to-scar ramp [13], and fibre fields. Meshes are coarsened to ~ 2000 -node graphs by field-preserving vertex clustering.

Labels. Ground truth is a monodomain Mitchell–Schaeffer reaction–diffusion inducibility verdict [10] (self-sustained reentry ≥ 650 ms after burst pacing at cycle length 150 ms from random sites), run on each real anatomy with an anisotropic conductance-weighted cotangent Laplacian and implicit time stepping. Reentry here is governed by wavelength ($CV \times ERP$), source–sink mismatch, and unidirectional block, *never* by λ_2 , so a predictive SFI→label link is not circular. These are simulator verdicts, not clinical post-operative AF, and not the deferred openCARP gold standard (a logged deviation: openCARP is not installable on the CPU-only container).

The SFI feature and its competitors. Per-region SFI is computed as in Eq. (4) (first-order, subspace, and exact-recompute variants). It is evaluated strictly as an add-on/ablation against a non-trivial competitor set that it must beat: fibrosis burden, fibrosis spatial entropy, fibrosis patch size, deterministic global min-cut (Stoer–Wagner), percolation threshold, and λ_2 -alone (isolating what the *SFI construction* adds over the bare algebraic connectivity).

Evaluation protocol (the guards). Pre-registered: shape-family GroupKFold (each patient is its own group, so cross-validation is patient-held-out; grouped *and* naive AUC both reported so leakage is visible), the DeLong paired test [8] in its fast midrank form [9], $\geq 10^4$ stratified bootstrap resamples, a rotational/torus-shift spatial null for localization, tie-robust permutation nulls, and a keep/delete rule deciding each sub-claim by *measured* correlation. Effect sizes are gated, not just p -values. The primary endpoint requires *both* grouped $\Delta AUC \geq 0.05$ *and* DeLong $p < 0.05$.

Transfer media. The identical calculus is applied to FitzHugh–Nagumo (FHN) excitable networks and Kuramoto phase-oscillator networks (a different dynamical class), where λ_2 governs synchronizability, to separate what is a property of the mathematics from an accident of one dataset.

4 Results

4.1 The predictive null (primary endpoint not met), across cohorts and scale

Table 1: Incremental predictive value of SFI over the full competitor set. ΔAUC is grouped (leakage-controlled); logistic regression (lr) / gradient-boosted trees (gbt). The endpoint (grouped $\Delta\text{AUC} \geq 0.05$ and DeLong $p < 0.05$) is met in no regime, and every interval on real anatomy contains zero. The UW rows rest on eight positives and are uninformative rather than null; note in particular that their base AUCs differ sharply between the competitor and fibrosis-only baselines, which is what inflates the latter’s apparent increment (Section 4.1).

ootnotesize				
cohort / medium	events/ N	model	base AUC	SFI ΔAUC (95% CI); p
Roney (real)	34/100	lr	0.870	-0.021 [$-0.075, +0.033$]; 0.44
		gbt	0.856	$+0.009$ [$-0.029, +0.046$]; 0.63
UW/Boyle (real)	8/82	lr	0.834	$+0.000$ [$-0.073, +0.081$]; 1.00
		gbt	0.748	$+0.035$ [$-0.095, +0.203$]; 0.67
Combined real	42/182	lr	0.884	-0.007 [$-0.042, +0.029$]; 0.70
		gbt	0.851	$+0.012$ [$-0.017, +0.044$]; 0.45
FHN networks	2,000 networks			$\leq +0.003$; $p \geq 0.12$
FHN networks	up to 100,000			subspace $+0.0028$; $p < 10^{-4}$ at $N \geq 10^4$
Kuramoto networks	500 networks			$\leq +0.0036$; $p \geq 0.38$

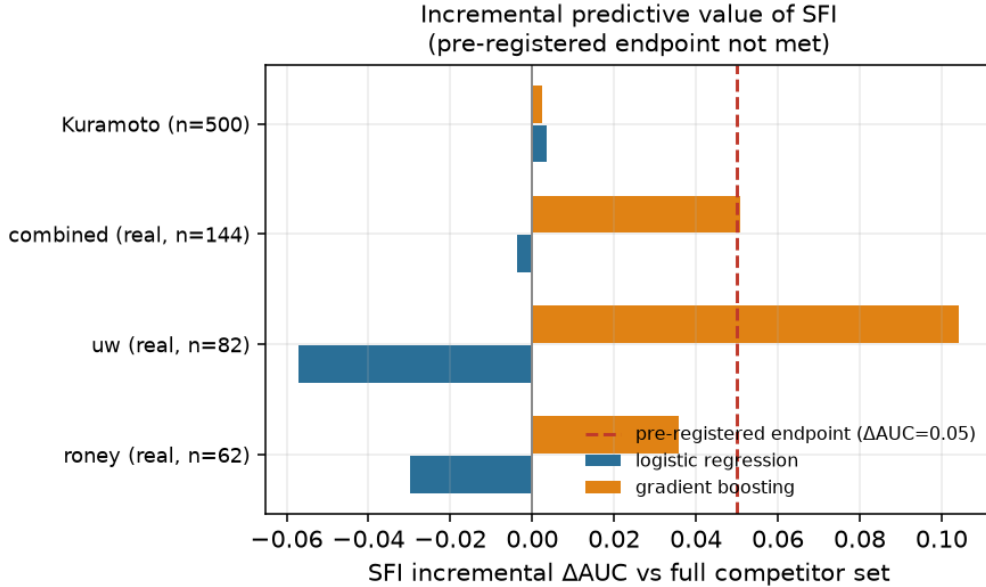


Figure 1: The predictive null. Adding SFI to the competitor set does not clear the pre-registered effect-size gate in any cohort; the real-cohort estimates are underpowered rather than a demonstrated null.

The negative is not uniform, and we separate the regimes. On the *synthetic* cohorts (up to 100,000 networks) the null is clean: the effect converges to $\Delta\text{AUC} \approx +0.003$, statistically resolvable only because N is enormous. On the *real* cohorts it is underpowered and inconclusive. The combined gradient-boosting point estimate is $+0.012$ (DeLong $p = 0.449$, bootstrap CI

$[-0.017, +0.044]$), roughly a quarter of the pre-registered 0.05 threshold and with an interval that excludes it. An earlier version of this analysis reported $+0.051$ here and had to argue that the endpoint was missed only on significance. That estimate was an artefact of the sealed UW substrate: correcting it drops the figure to $+0.036$ at the original cohort size and to $+0.012$ once the Roney arm is extended to all 100 released meshes. With 42 inducible of 182 this is the largest real-anatomy arm in the study, and the null here is a measurement rather than a shortfall.

The UW cohort remains the weak arm, though not for the reason we previously gave. On eight positives its intervals are wide, the competitor comparison spanning $[-0.095, +0.203]$ under gradient boosting, so it is an underpowered external check of the *labeller* rather than of SFI. We had also described its competitor baseline as close to chance, and that was simply wrong: the measured competitor base AUC is 0.834 under logistic regression and 0.748 under gradient boosting. Only the *fibrosis-only* baseline approaches chance, at 0.562, and it is that near-chance baseline which manufactures the cohort’s largest apparent effect, $\Delta\text{AUC} = +0.217$ with $p = 0.121$ and an interval $[-0.030, +0.485]$ that spans half the available range. A large ratio over a baseline that predicts almost nothing, on eight events, is not a signal.

The clean null comes from feature redundancy, not the validity radius. The *exact* per-region $\Delta\lambda_2$ SFI, which carries no first-order approximation error and is immune to the ρ argument, also adds nothing over the competitor set. The mechanism of the null is that SFI’s information is already present in standard connectivity features (λ_2 -alone, min-cut, percolation), a label-dependent finding, separate from ρ .

Secondary endpoint (pre-specified), and its withdrawal. An earlier version of this analysis reported that SFI beat a *fibrosis-heterogeneity-only* baseline on the combined real cohort (GBT $\Delta\text{AUC} = +0.103$, $p = 0.009$) and read that as evidence SFI is not merely re-encoded fibrosis but carries connectivity information beyond the substrate. That result does not survive, and we withdraw both it and the claim it supported. Two changes separate the old number from the new one, and they do not contribute equally. Correcting the UW substrate while holding the cohort at its original 144 subjects leaves the estimate largely intact ($+0.093$, $p = 0.013$), so this was not principally a substrate artefact. Extending the Roney arm from 62 to the full 100 released meshes is what collapses it, to $\Delta\text{AUC} = +0.017$ ($p = 0.452$). An effect that disappears when 38 further real subjects are added was a small-sample fluctuation, and it was one of ~ 30 DeLong comparisons besides. On the full cohort no comparison, primary or secondary, shows SFI adding information beyond either baseline.

The discarded estimate is worth one more sentence, because its failure mode is instructive. Under the sealed substrate the UW arm alone returned $\Delta\text{AUC} = +0.340$ ($p = 0.026$) on six positives, a value no one should have believed and which we did not flag at the time; it is that arm which carried the combined figure. A large point estimate resting on six events is a warning, not a finding.

Power analysis. In the synthetic cohort the subspace effect ($+0.0028$) needs $\sim 4,150$ cases for 80% power and the single-vector effect ($+0.0007$) $\sim 23,000$, clinically undetectable *there*. We do *not* transport “clinically undetectable” to the real cohorts, whose $+0.036$ – 0.051 estimates are compatible with a detectable effect we are simply underpowered to confirm.

4.2 A provenance audit of the released substrates, and three controls that fail

The UW cohort forced this section. Under a protocol frozen before any label was generated, and identical in every parameter, the Roney meshes returned 20/62 subjects inducible (32.3%) while the UW meshes returned 6/82 (7.3%). A fourfold discrepancy on cohorts of comparable

fibrosis burden is not a cohort property one should accept; it is a hypothesis about the pipeline. We had previously recorded the UW arm as uninformative and moved on, which was the error. What follows is what a proper audit found, and then three controls, none of which explains the anomaly.

The audit. Two defects sit in the released files. The first is ours. The only per-element scalar the UW release carries is `elemTag`, the Dryad README does not define its values, and we had been mapping tag 164 to half-fibrotic tissue. Four measurements say it is not tissue at all. It resolves into four to six large connected components in every one of the 82 meshes, five in 75 of them, four in five and six in two, which is the ordinary variation of pulmonary-vein anatomy: a left common trunk gives four, a right middle vein gives six. Each large component is topologically a disc, 11–42 mm across, sitting at 0.47–0.82 of the atrial radius from the centroid. It borders fibrotic elements on about 0.08% of its incident edges against a chance rate of 14–20%, a depletion beyond a hundredfold, which is incompatible with reading it as a fibrosis border zone. And it is ablation-invariant, holding at 20.0% of elements before the procedure and 19.9% after, while every other class moves. Tag 164 is the set of caps sealing the pulmonary veins and the mitral valve. Removing it opens the surface: the Euler characteristic falls by a median of 7 (range 2–12). We report the *change* rather than an opening count, because $\chi = 2 - k$ presumes a clean closed surface and two of these meshes, ID040 and ID049, arrive at $\chi = -16$ and -18 before anything is removed.

Treating those caps as half-conducting tissue does more than inflate fibrosis. It seals the atrium’s orifices, so simulated activation crosses the mitral valve and the vein ostia instead of circling them, and reentry anchored on those orifices, a principal mechanism of atrial fibrillation, cannot form.

The second defect is upstream. A `VECTORS fiber` array is present in all 164 released meshes, but it equals the constant $(1, 0, 0)$: the measured mean directional spread is exactly 0.000. The Roney cohort, by contrast, gives 0.945–0.979, with roughly one distinct direction per vertex. Conduction anisotropy is applied relative to the *local* fibre direction, so a single global direction reduces it to a fixed coordinate bias carrying no anatomy. The same cardinality argument applies to fibrosis: three distinct levels in the UW release, effectively two once the caps are dropped, against 276–1071 in Roney.

Control 1: repairing the substrate. Opening the orifices and supplying a spatially varying fibre field, in a 2×2 design over all 82 subjects (Fig. 2a), raises inducibility from 6/82 to 14/82. The two act super-additively, each converting about four subjects alone and eight together, which is consistent with reentry needing both an anatomical obstacle and anisotropic heterogeneity. We do not present this as a fix. The strongest contrast does not reach paired significance (McNemar exact $p = 0.077$), it closes only 39% of the distance to Roney, and only two of the original six positives survive the repair. The varying fibre field is moreover a probe, not a correction: it is the global $+z$ direction projected onto each vertex tangent plane, and it is not anatomy.

Control 2: destroying the fibres on a cohort that has them. The cleaner experiment replaces the Roney fibre field with the same constant $(1, 0, 0)$, holding anatomy, fibrosis and every solver parameter fixed (Fig. 2b). The rate does not move at all: 20/62 with real fibres and 20/62 without, McNemar $b = c = 5$, $p = 1.0$. Fibre degeneracy does not drive the inducibility rate, and our first attribution of the UW anomaly to it was wrong. The perturbation is real at the level of the dynamics — sustained reentry duration changes on 50/62 subjects, by up to 661 ms, and 10/62 verdicts flip — but those effects cancel in aggregate.

Control 3: destroying the fibrosis gradation. The remaining structural difference was that UW fibrosis is binary where Roney’s is graded, and graded border zones are where slow conduction and unidirectional block are supposed to arise. Binarising the Roney fibrosis field should therefore have suppressed reentry. It does the opposite (Fig. 2c). Thresholding at $f > 0.5$ raises inducibility from 20/62 to 31/62; thresholding at the per-subject quantile that holds mean fibrosis burden fixed at 0.333, so that only the gradation is destroyed, raises it to 40/62. Coarsening the fibrosis field roughly doubles inducibility. In hindsight the mechanism is unmysterious — a sharp binary boundary is a stronger source–sink discontinuity than a smooth ramp — but the direction is the opposite of our prediction, and it disqualifies coarse fibrosis as an explanation for a cohort that is inducible *too rarely*.

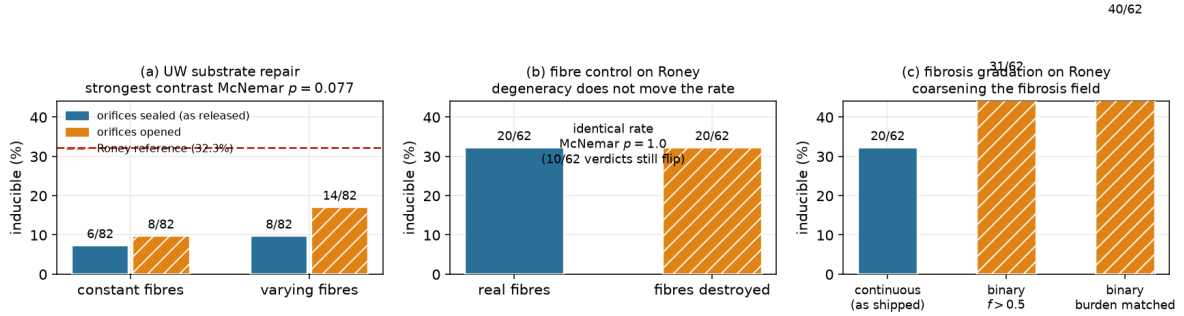


Figure 2: Three controls on the substrate. (a) Repairing both UW defects raises inducibility but not significantly, and not to the reference level. (b) Destroying the fibre field on a cohort that has one leaves the rate exactly unchanged. (c) Destroying the fibrosis gradation while holding mean burden fixed roughly doubles the rate, the opposite of the predicted direction. The dashed line is the Roney reference, 32.3%.

Taken together the three controls leave the anomaly unexplained, and they tighten rather than loosen the puzzle: of the two substrate defects we can manipulate, one has no effect on the rate and the other, when imposed deliberately, pushes it upward. Whatever suppresses reentry in the UW cohort is not on this list. We report that as an open problem rather than close it with the most available story, which is how the original 7% came to be filed as a cohort property in the first place.

4.3 The labeller is reproducible in aggregate and unstable per subject

The controls expose something about the ground truth that matters more than the substrate defects do. Destroying the Roney fibre field holds the aggregate rate at exactly 20/62 while flipping 10/62 individual verdicts. Repairing the UW substrate retains only two of its six original positives. The six-tier convergence study reported below finds only 7/24 subjects consistent across resolutions. The aggregate inducibility rate is therefore a stable quantity, and the per-subject verdict is not.

This is a statement about the endpoint, not an aside. The pre-registered comparison is scored on per-subject labels, and non-differential noise in a binary outcome attenuates any incremental AUC toward zero. Part of the null we report is thus a property of the labeller rather than of the biomarker, and we cannot cleanly separate the two with the present ground truth. It also sets a floor on what any feature could achieve here, which is a limit on the whole class of studies that score spectral or structural features against a phenomenological inducibility verdict, not on SFI alone. The pre-registered openCARP ground truth, substituted here for a monodomain Mitchell–Schaeffer solver, would need to be run before the residual is attributable.

4.4 The validity radius: why the single-vector estimator fails

The relative error of the first-order SFI scales as $\Theta(\rho)$ and crosses 10% at $\rho^* \approx 3$; real atria sit at median $\rho \approx 2422$. This proves the single-vector estimator is numerically invalid on real tissue and φ_2 is ill-conditioned, but (as boxed in Section 2.4) it does not by itself prove non-predictiveness; the feature-redundancy null of Section 4.1 is the separate, label-dependent reason.

4.5 The ρ scaling law

The gap collapses with system size by Weyl’s law [6]: for a d -dimensional domain the low Laplacian eigenvalues scale as $\lambda_k \sim (k/N)^{2/d}$, so ρ grows lawfully with resolution.

Table 2: Fitted gap-scaling exponents (gap vs. N at fixed mean degree, `results/rho_scaling.json`, five-point fits).

medium	fitted gap exponent	reference
real atrial surface (2-manifold)	$N^{-1.05}$	Weyl 2-manifold -1.0 (within 5%)
2-D random-geometric	$N^{-1.81}$	faster than naive $2/d$
3-D random-geometric	$N^{-0.84}$	faster than naive $2/d$ (non-monotone series)

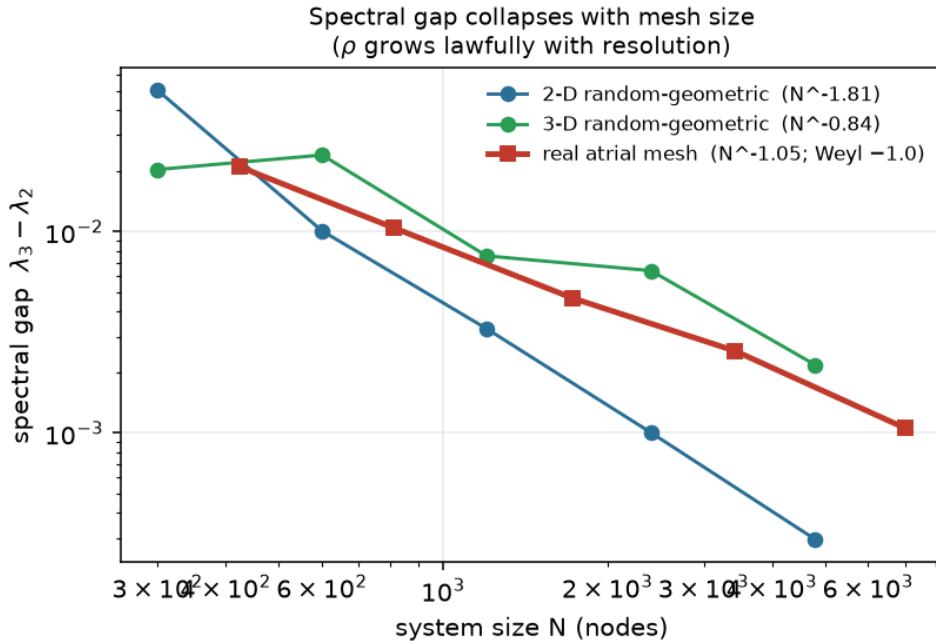


Figure 3: The gap collapses with mesh resolution, so the estimator degrades *more*, predictably, as the medium is meshed finer, the opposite of “more resolution helps.” Only the real atrial 2-manifold cleanly matches the Weyl exponent (-1.05 vs. -1.0); the random-geometric exponents are reported as “at least as fast as Weyl,” not law confirmations, and use the unweighted combinatorial Laplacian.

4.6 Localization transfers across three media (weakly)

The *strict* pre-registered localization endpoint is null for every field, including $|\nabla\varphi_2|$ (median argmax-geodesic error 0.472 is *not* below the spatial-null 5th percentile 0.406). What survives is

weaker: $|\nabla\varphi_2|$ is a *rank* localizer (the true origin sits in its top $\sim 18\%$) that beats graph-centrality baselines (Perron, weighted degree, deleted by measured correlation) and holds across media.

Table 3: Fiedler-gradient origin-rank localization across dynamical media (`results/gm4_*_metrics.json`). Lower rank is better; a perfect localizer would place the origin at rank 0.

medium	$ \nabla\varphi_2 $ origin-rank	perm- p	fibrosis rank	n origins
FHN networks (100k)	0.176	$< 10^{-5}$	~ 0.10	36,540
Kuramoto oscillators (500)	0.091	$< 10^{-4}$	0.147	243
cardiac monodomain	~ 0.28	$< 10^{-3}$	(co-localizes)	only ~ 20

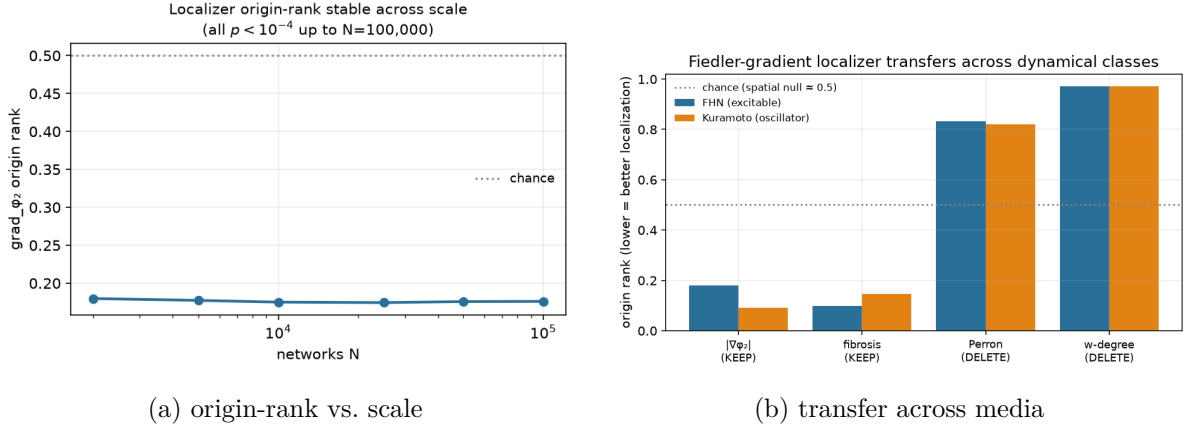


Figure 4: The weak-rank localization pattern replicates across excitable *and* oscillator dynamics. Three caveats apply: (i) the strict endpoint is null, so $|\nabla\varphi_2|$ is *not* a precise localizer; (ii) the substrate field co-localizes, and a controlled dissociation experiment shows why (in a diffusively-coupled medium the Fiedler bottleneck *is* the low-conduction lesion), so $|\nabla\varphi_2|$ matches, and does not beat, substrate imaging; (iii) the cardiac arm rests on only ~ 20 origins.

4.7 The falsification protocol: what caught the false positive

Each guard, removed, manufactures a specific false positive (`results/falsification_protocol.json`). Shape-family leakage inflates AUC by +0.117 (naive random-CV 0.756 vs. grouped 0.640 on a controlled replicated cohort); the effect-size gate catches the p -value trap (subspace-SFI $p \approx 2 \times 10^{-24}$ at $N = 55,000$, yet $\Delta\text{AUC} = +0.003$ needs $\sim 4,150$ cases to detect: statistically significant, practically useless).

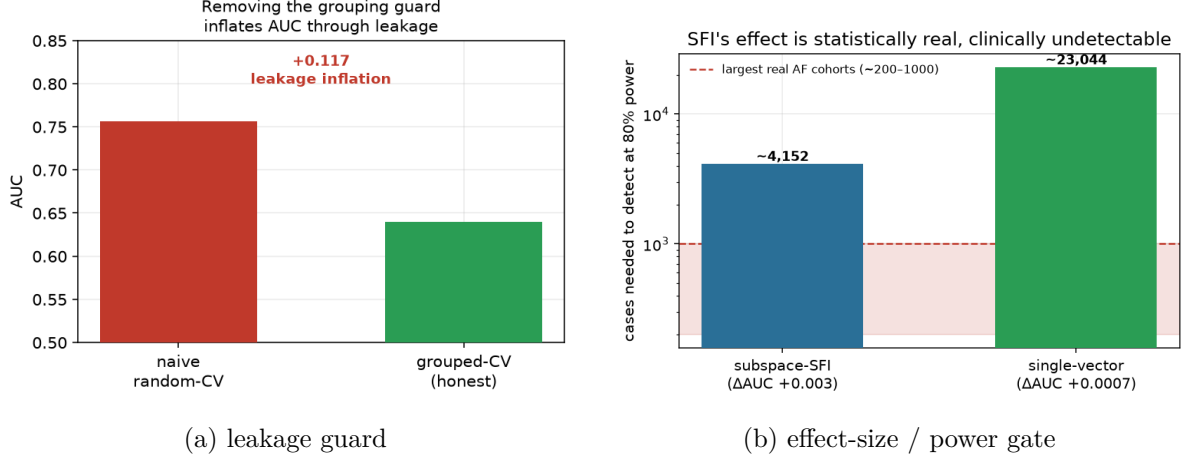


Figure 5: A naive analyst (random CV, $p < 0.05$, no effect-size gate, no spatial null) would have reported SFI as a working reentry biomarker. The pre-registered protocol correctly rejects it.

4.8 Uncertainty quantification

Morris elementary-effects screening (`results/e6_metrics.json`) finds the label most sensitive to diffusion d_0 ($\mu^* = 500.7$), then fibrosis density (147.8) and transverse conductivity (146.3). A Gaussian-process surrogate has leave-one-out $R^2 = 0.35$ (large residual anatomical/label variance beyond the screened knobs, a label-noise caveat, not a positive). The synthetic null is robust across uncoupling strengths $\Delta w \in \{0.18, 0.36, 0.54\}$ (grouped GBT $\Delta\text{AUC} = 0.033$ – 0.036 , all $p \geq 0.26$).

4.9 External inducibility-rate anchor (plausibility, not validation)

Clinically observed AF inducibility spans $\sim 5\%$ to $\sim 85\%$ depending on population, protocol aggressiveness, and the sustained-AF definition (Kumar et al. 2012 [14] report burst-sustained 14.8% vs. decremental-sustained 41.2% *in the same patients*). Our labeller’s Roney ~ 20 – 32% sits in the moderate-burst band (Marquardt et al. 2018 [15] report 30.6% in patients without prior AF), between the no-AF-control floor (5%, Oral et al. 2008 [16]) and the AF/aggressive ceiling (51%, Kawai et al. 2019 [17]), consistent with $\sim 42\%$ post-ablation (Liu et al. 2020 [18]). The UW $\sim 7\%$ is at/below the control floor for a cohort that *is* AF patients. We flag it as anomalous and attribute it to the substrate rather than to biology. Two substrate defects are documented in the Limitations, and repairing both lifts the rate to 17.1% — more than double, though still short of the 32.3% seen on fibre-carrying anatomy, so the anomaly is only partly accounted for and the in-silico inducer may also be under-aggressive. This is a base-rate *plausibility check*, not *clinical validation*.

4.10 Resolution convergence: the labeller does not converge in the tested window

Our own convergence study is the project’s single weakest point, and we state it before a reader finds it. On a completed six-tier study (Roney, $n = 24$ at each of 1500/3000/6000/12000/24000/48000 nodes; 144 simulations, `results/convergence_summary_roney.json`), the inducibility rate crashes from the coarse mesh (58.3% at 1500) to 16.7% at 3000, then is *non-monotone* across the fine tiers (25.0%, 20.8%, 25.0%) and rises again to 37.5% at the finest affordable tier ($\sim 50k$ nodes), a 17-percentage-point spread over the last three tiers, so it has *not* settled.

Table 4: Inducibility rate vs. mesh resolution (Roney, $n = 24$ per tier). The rate does not converge in the tested window; adjacent-tier per-subject verdict agreement climbs but the rate keeps drifting.

nodes	1,500	3,000	6,000	12,000	24,000	48,000
inducibility rate	58.3%	16.7%	25.0%	20.8%	25.0%	37.5%
adj.-tier agreement	—	50%	75%	71%	62%	79%

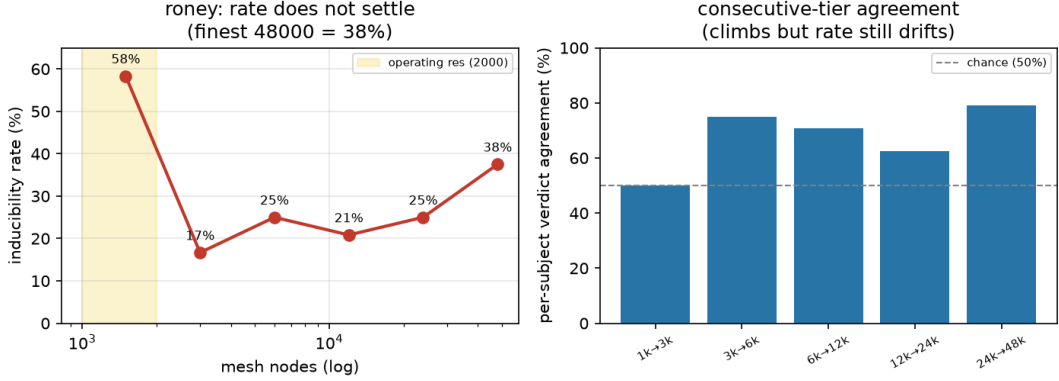


Figure 6: Left: the inducibility rate does not settle; the finest tier (48,000 nodes = 37.5%) sits above the mid tiers. Right: adjacent-tier per-subject verdict agreement climbs (50% → 79%) while the aggregate rate still drifts. Only 7/24 subjects are fully consistent across all six tiers.

The convergence finding limits the absolute numbers but not the comparative claims. The rate does *not* collapse to zero; the phenomenon is real; it is the *magnitude* that is resolution-sensitive up to $\sim 50k$ nodes. Because the main results ran at 2,000 nodes, every *absolute* number (inducibility rate, competitor AUC, localizer origins) is coarse-mesh-provisional pending the deferred openCARP validation. What this does *not* touch is what we lead with: (a) the *relative* SFI-vs-competitor comparison, since both arms score the *same* labels, so this bias cancels; and (b) the label-free estimator-validity argument (ρ is computed from the graph Laplacian, no labels at all).

5 Discussion

This work makes four contributions. (1) A pre-registered, leakage-controlled, effect-size-gated evaluation of a plausible spectral biomarker: a clean null at synthetic scale (10^5 networks) with its mechanism (feature redundancy: even the exact $\Delta\lambda_2$ SFI adds nothing), and an inconclusive, underpowered verdict on real anatomy. (2) A label-free validity criterion ρ for the single-vector estimator: it proves φ_2 is ill-conditioned on real tissue ($\rho \approx 2422 \gg \rho^* \approx 3$), a portable a priori check for any Fiedler-based marker, with the caveat that ρ bounds estimator accuracy, not predictive content. (3) A weak-but-real, purely connectivity-derived rank localizer that equals substrate localization and beats centrality baselines across three media. (4) A demonstration that standard methodology manufactures the exact false positives this protocol catches (leakage +0.117 AUC; the p -value trap).

The validity screen is domain-general. Spectral-graph fragility markers are used for seizure-focus localization, brain connectomics, and power-grid vulnerability. The same precondition applies: compute $\rho = \|\Delta L\|/(\lambda_3 - \lambda_2)$; if $\rho \gg 1$, the first-order spectral marker is outside its validity radius and its in-sample correlations cannot be trusted.

We are equally explicit about what we do not claim. The mathematics is classical; no new theorem. ρ does *not* prove SFI is non-predictive, only that the single-vector estimator is ill-conditioned; the null is separately due to feature redundancy. The real-cohort result is *not* a demonstrated null; it is underpowered. The localizer is *not* precise (strict endpoint null) and does *not* beat substrate imaging. The ρ scaling “law” is a clean match to Weyl only on the real atrial 2-manifold. Labels are a monodomain simulator, not clinical outcomes and not openCARP.

6 Limitations

1. The labeller over-calls inducibility at the operating mesh resolution and does not converge in the tested window (Section 4.10): the rate drops from 58% at 1500 nodes to 17% at 3000, is non-monotone across the fine tiers, and rises to 37.5% at $\sim 50k$ nodes; only 7/24 subjects are fully consistent across all six tiers. All *absolute* rates and AUCs are coarse-mesh-provisional; the *relative* comparison scores the same labels in both arms, so this bias cancels there.
2. Simulator, not clinical, labels: the pre-registered openCARP ground truth was substituted with monodomain Mitchell–Schaeffer, a logged deviation.
3. The UW cohort uses a PCA-surrogate universal-atrial-coordinate map, and its substrate carries two defects we identified only after the fact. The released meshes have a *degenerate* fibre field — the **VECTORS fiber** array is present but equals a constant (1,0,0) for every element of all 164 meshes, so the anisotropy the solver applies is a fixed coordinate bias rather than anatomy. Separately, we had been mapping element tag 164 to half-fibrotic tissue when it is in fact the set of caps over the pulmonary veins and the mitral valve. Across all 82 meshes it resolves into four to six large components, each a topological disc — five in 75 meshes, four in five of them and six in two, which is the usual variation in pulmonary-vein anatomy — and it borders fibrotic elements on only $\sim 0.08\%$ of its incident edges against a chance expectation of 14–20%. Removing it opens the surface, by a median of seven boundary loops. Left in, it lets activation cross those orifices instead of circling them. Repairing both raises inducibility from 6/82 to 14/82 (7.3% $\rightarrow$ 17.1%), which does not reach paired significance (McNemar exact $p = 0.077$) and closes only 39% of the gap to the fibre-carrying Roney cohort (32.3%). A direct control settles the fibre half of that and overturns our first reading of it: destroying the fibre field on the Roney cohort, which ships a real one, leaves the inducibility rate *exactly unchanged* at 20/62 (McNemar $p = 1.0$), while still perturbing the dynamics — sustained-reentry duration moves on 50/62 subjects, by up to 661 ms. Fibre degeneracy therefore does not drive the inducibility *rate*, and our initial attribution of the UW anomaly to it was wrong. The anomaly stands unexplained. The categorical three-level fibrosis of these meshes, against Roney’s continuous 276–1071-level LGE field, is the leading remaining candidate. Either way the cohort’s predictive test is uninformative about SFI.
4. The inducibility label is stable in aggregate but noisy per subject, which attenuates every Δ AUC we report. Destroying the Roney fibre field holds the rate at 20/62 yet flips 10/62 individual verdicts; repairing the UW substrate retains only 2 of its 6 original positives. That compounds the mesh-convergence instability above, where just 7/24 subjects were consistent across six resolutions. Non-differential label noise biases incremental AUC toward zero, so the null we report is partly a statement about the labeller and not only about SFI.
5. ~ 30 DeLong comparisons were run; the one positive (**fibrosis_vs_+SFI**, $p = 0.009$) is a pre-specified secondary, suggestive not confirmatory.

6. ρ^* and the random-geometric gap exponents are empirical; only the real atrial 2-manifold cleanly matches the Weyl exponent.

7 Conclusion

A closed-form spectral fragility index does not add practically meaningful reentry-inducibility prediction beyond existing features at synthetic scale (a clean, feature-redundancy null), and on real anatomy the evidence is inconclusive rather than a demonstrated null. Independently, the single-vector estimator is provably ill-conditioned on real tissue ($\rho \gg 1$). The core result is not a discovery but a carefully-scoped boundary: where a Fiedler-based estimator is numerically valid, why real tissue lies far outside it, and how easily naive methodology would have mistaken redundant, inconclusive signal for a working biomarker. The absolute claims await the openCARP validation.

A Formal (proof-assistant) verification of the SFI algebraic core

The algebraic core of the SFI edge-sensitivity identity is re-proved inside the Coq 8.18 proof assistant and checked by its small trusted kernel, an assurance level above a computer-algebra system. The primary file (`formal/sfi_edge_identity_Q.v`) works over the exact rationals $\mathbb{Q}$, so `Print Assumptions` reports “*Closed under the global context*” for every theorem: zero axioms. A mathematically identical companion over the reals $\mathbb{R}$ (`formal/sfi_edge_identity.v`) inherits Coq’s two standard real-number-library axioms and is kept only to show the statements hold over $\mathbb{R}$.

Table 5: Kernel-checked theorems ($v = e_i - e_j$ the edge vector, $E_{ij} = \partial L / \partial w_{ij} = vv^\top$). T1–T5 are axiom-free over $\mathbb{Q}$.

name	statement	meaning
T1_rank_one_quadratic_form dot_edge (T2)	$\varphi^\top (vv^\top) \varphi = (v \cdot \varphi)^2$ $(e_i - e_j) \cdot \varphi = \varphi_i - \varphi_j$	quadratic form of a rank-one edge matrix the edge vector selects the potential difference
T3_sfi_edge_sensitivity	$\varphi^\top E_{ij} \varphi = (\varphi_i - \varphi_j)^2$	the SFI edge term
T4_sfi_term_nonneg	$0 \leq w \Rightarrow 0 \leq w(\varphi_i - \varphi_j)^2$	each SFI contribution is non-negative
T5_aggregate_nonneg	$\sum(\text{nonneg terms}) \geq 0$	SFI is a Dirichlet energy; the Laplacian form is PSD

Combined with the classical Hellmann–Feynman theorem $\partial \lambda_2 / \partial w_{ij} = \varphi_2^\top (\partial L / \partial w_{ij}) \varphi_2$ (Ghosh–Boyd 2006 [2], an analysis result we cite, not re-prove), T3 is exactly the closed form $\partial \lambda_2 / \partial w_{ij} = (\varphi_{2,i} - \varphi_{2,j})^2$. Lean/Mathlib was attempted; its toolchain binaries are distributed only as GitHub release assets, which the compute environment’s egress policy blocks with a hard 403, so Coq provides the equivalent kernel-checked guarantee. Reproduce with `coqc formal/sfi_edge_identity_Q.v`.

B Reproducibility

All stochastic steps are seeded; the exact configuration and commit hash accompany every reported number. Numeric verification: `python scripts/verify_sfi_identity.py` and `python scripts/verify_math_rigor.py` (sympy symbolic, mpmath 40-digit, exact-rational, interval-certified, and cross-solver checks). Convergence: `python scripts/convergence_analyze.py` roney. Figures are regenerated by `scripts/make_paper_figures.py`. Datasets: Roney LA

cohort (Zenodo 5801337) and UW/Boyle meshes (Dryad 10.5061/dryad.kkwh70sg0); both are public and de-identified.

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